University of Ottawa

Department of Mathematics and Statistics

MAT 2362 : Foundations of Mathematics

Professor: Hadi Salmasian

Final Exam

December 19, 2023

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 180 minutes to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

(g) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. **Phones and devices must be turned off and put away in your bag**. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave the exam immediately, and academic fraud allegations will be filed, which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you are complying with the above statement.

Signature _____

Please do not write in the table below.

Question	1	2	3	4	5	6	7	8	9	10	Total
Maximum	5	5	7	4	3	4	4	7	4	2	45
Grade											

1. (b) **[2 points]** Write down the negation of the following quantifier sentence:

$$\forall_{x\in\mathbb{R}}\exists_{y\in\mathbb{R}}\forall_{z\in\mathbb{R}}\big((z^2+zy+x^3\geq 0)\wedge(y>x)\big)$$

Solution:

$$\exists_{x\in\mathbb{R}}\forall_{y\in\mathbb{R}}\exists_{z\in\mathbb{R}}\big(z^2+zy+x^3<0\ \vee\ y\leq x\big)$$

(b) **[3 points]** Prove the following equivalence algebraically. You should *not* use truth tables. Justify each step by writing the number of the equivalence you are using from the table below.

$$P\vee((Q\vee P)\Rightarrow P)\equiv P\vee\neg Q$$

Solution:

$$\begin{aligned}
P\vee((Q\vee P)\Rightarrow P) &\equiv P\vee(P\vee\neg(Q\vee P)) && (18),\ (1)\\
&\equiv P\vee((P\vee\neg Q)\wedge(P\vee\neg P)) && (14),\ (17)\\
&\equiv P\vee((P\vee\neg Q)\wedge T) && (12)\\
&\equiv P\vee(P\vee\neg Q) && (8)\\
&\equiv (P\vee P)\vee\neg Q && (3)\\
&\equiv P\vee\neg Q && (5).
\end{aligned}$$

	Equivalence			Name or comment
(1)	$P\vee Q$	$\equiv$	$Q\vee P$	Commutativity of $\vee$
(2)	$P\wedge Q$	$\equiv$	$Q\wedge P$	Commutativity of $\wedge$
(3)	$(P\vee Q)\vee R$	$\equiv$	$P\vee(Q\vee R)$	Associativity of $\vee$
(4)	$(P\wedge Q)\wedge R$	$\equiv$	$P\wedge(Q\wedge R)$	Associativity of $\wedge$
(5)	$P\vee P$	$\equiv$	P	Idempotence
(6)	$P\wedge P$	$\equiv$	P	Idempotence
(7)	$P\vee\mathbf{F}$	$\equiv$	P	$\mathbf{F}$ is neutral for $\vee$
(8)	$P\wedge\mathbf{T}$	$\equiv$	P	$\mathbf{T}$ is neutral for $\wedge$
(9)	$P\vee\mathbf{T}$	$\equiv$	$\mathbf{T}$	$\mathbf{T}$ is absorbent for $\vee$
(10)	$P\wedge\mathbf{F}$	$\equiv$	$\mathbf{F}$	$\mathbf{F}$ is absorbent for $\wedge$
(11)	$\neg\neg P$	$\equiv$	P	Double negation
(12)	$P\vee\neg P$	$\equiv$	$\mathbf{T}$	Excluded middle
(13)	$P\wedge\neg P$	$\equiv$	$\mathbf{F}$	Contradiction
(14)	$\neg(P\vee Q)$	$\equiv$	$\neg P\wedge\neg Q$	De Morgan's law
(15)	$\neg(P\wedge Q)$	$\equiv$	$\neg P\vee\neg Q$	De Morgan's law
(16)	$P\wedge(Q\vee R)$	$\equiv$	$(P\wedge Q)\vee(P\wedge R)$	Distributivity of $\wedge$ over $\vee$
(17)	$P\vee(Q\wedge R)$	$\equiv$	$(P\vee Q)\wedge(P\vee R)$	Distributivity of $\vee$ over $\wedge$
(18)	$P\Rightarrow Q$	$\equiv$	$\neg P\vee Q$	
(19)	$P\Leftrightarrow Q$	$\equiv$	$(P\Rightarrow Q)\wedge(Q\Rightarrow P)$	
(20)	$P\Leftrightarrow Q$	$\equiv$	$(P\wedge Q)\vee(\neg P\wedge\neg Q)$	

2. **[5 points]** Consider a sequence $a_0, a_1, a_2, a_3, \ldots$ of real numbers. Write down the following sentences using quantifiers. In your answers you can only use the following mathematical symbols:

$$\forall\ ,\ \exists\ ,\ \vee\ ,\ \wedge\ ,\ \neg\ ,\ \Rightarrow\ ,\ \leq\ ,\ <\ ,\ \geq\ ,\ >\ ,\ =\ ,\ \neq\ ,\ +\ ,\ -\ ,\ \times\ ,\ \div$$

$$\mathbb{N}\ ,\ \mathbb{Z}\ ,\ \mathbb{Q}\ ,\ \mathbb{R}\ ,\ \in\ ,\ \notin\ ,\ \text{ any real number }\ ,\ \text{ any variable (e.g., } m, n, \ldots).$$

(a) All of the a_n are positive real numbers.

Solution: $\forall_{n\in\mathbb{N}}(a_n > 0)$.

(b) All of the a_n for n even are irrational numbers.

Solution: $\forall_{n\in\mathbb{N}}\big((\exists_{k\in\mathbb{Z}} n = 2k) \Rightarrow (a_n \in \mathbb{R}\ \wedge\ a_n \notin \mathbb{Q})\big)$

(c) The minimum value among the a_n occurs for $n = 25$.

Solution: $\forall_{n\in\mathbb{N}}(a_n \geq a_{25})$

(d) We have $a_n = 0$ for some odd n.

Solution: $\exists_{n\in\mathbb{N}}(\exists_{k\in\mathbb{N}}(n = 2k+1) \wedge (a_n = 0))$

(e) The numbers a_n are pairwise distinct.

Solution: $\forall_{m\in\mathbb{N}}\forall_{n\in\mathbb{N}}(m \neq n\ \Rightarrow\ a_m \neq a_n)$

(f) Only finitely many of the a_n are nonzero.

Solution: $\exists_{m\in\mathbb{N}}\forall_{n\in\mathbb{N}}(n > m \Rightarrow a_n = 0)$

3. (a) **[3 points]** Let $f : \mathbb{R} \to \mathbb{R} \times \mathbb{R}$ be the function defined by $f(t) = (t, t^2)$. Compute $f^{-1}(A)$ explicitly, where $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} \ : \ 2x + y \leq 3\}$. You should justify your answer.

Solution: We have $f^{-1}(t) = \{t \in \mathbb{R} \ : \ f(t) \in A\}$. Now

$$f(t) \in A \Leftrightarrow (t, t^2) \in A \Leftrightarrow 2t + t^2 \leq 3 \Leftrightarrow (t+1)^2 \leq 4.$$

Thus

$$f^{-1}(A) = \{t \in \mathbb{R} \ : \ (t+1)^2 \leq 4\} = \{t \in \mathbb{R} \ : \ |t+1| \leq 2\} = [-3, 1].$$

(b) **[1 point]** Write down the definition of a strict partial order.

Solution: A relation that is transitive and irreflexive is called a strict partial order.

(c) **[3 points]** Consider the relation on $\mathbb{R} \times \mathbb{R}$ defined by

$$(x, y) \lhd (x', y') \text{ if and only if } x - x' < 2y' - 2y.$$

Prove that $\lhd$ is a strict partial order.

Solution: Irreflexivity: assume that $(x, y) \lhd (x, y)$. Then $0 = x - x < 2y - 2y = 0$, i.e., $0 < 0$ which is a contradiction.

Transitivity: assume that $(x, y) \lhd (x', y')$ and $(x', y') \lhd (x'', y'')$. Then

$$x - x' < 2y' - 2y \ \wedge x' - x'' < 2y'' - 2y'.$$

Adding both sides of inequalities we obtain

$$(x - x') + (x' - x'') < (2y' - 2y) + (2y'' - 2y'),$$

hence $x - x'' < 2y'' - 2y$. Hence $(x, y) \lhd (x'', y'')$.

4. Let $f : X \to Y$ be a function from a set X to a set Y. Suppose that Y is equipped with an equivalence relation $\sim_Y$. We define the relation $\sim_X$ on X as follows:

$$\forall_{x,x' \in X}(x \sim_X x' \Leftrightarrow [f(x)] = [f(x')]),$$

where $[f(x)]$ for $x \in X$ denotes the equivalence class of $f(x)$ with respect to $\sim_Y$.

(a) **[3 points]** Prove that $\sim_X$ is an equivalence relation on X.

Solution: Reflexive:
Given $x \in X$ we have $f(x) = f(x)$, hence $[f(x)] = [f(x)]$ and thus $x \sim_X x$.
Symmetric: Given $x, x' \in X$ we have

$$x \sim_X x' \Rightarrow [f(x)] = [f(x')] \Rightarrow [f(x')] = [f(x)] \Rightarrow x' \sim_X x.$$

Transitive: for all $x_1, x_2, x_3 \in X$, if $x_1 \sim_X x_2$ and $x_2 \sim_X x_3$ then we have

$$[f(x_1)] = [f(x_2)] \ \wedge \ [f(x_2)] = [f(x_3)],$$

hence $[f(x_1)] = [f(x_3)]$. It follows that $x_1 \sim_X x_3$.

(b) **[1 point]** Suppose that $X = Y = \mathbb{R}$ and $\sim_Y$ is the equality on Y, i.e.,

$$\forall_{y,y' \in Y}(y \sim_Y y' \Leftrightarrow y = y').$$

Furthermore, assume that $f(t) = t^2 + 1$ for all $t \in \mathbb{R}$. Compute (explicitly) the equivalence class of $1 \in X$ with respect to the relation $\sim_X$ of part (a).

Solution: Given $t \in \mathbb{R} = X$, we have $t \sim_X 1$ if and only if $t^2 + 1 = f(1) = 2$, i.e., $t^2 = \pm 1$. Thus $[1] = \{1, -1\}$.

5. **[3 points]** Prove that for every 3 sets A, B, and C we have

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

Solution:

$$\begin{aligned}
x \in A \setminus (B \cup C) &\Leftrightarrow x \in A \ \wedge \ x \notin (B \cup C) \\
&\Leftrightarrow x \in A \wedge \neg(x \in B \cup C) \\
&\Leftrightarrow x \in A \wedge \neg(x \in B \vee x \in C) \\
&\Leftrightarrow x \in A \wedge ((x \notin B) \wedge (x \notin C)) \\
&\Leftrightarrow x \in A \wedge x \in A \wedge x \notin B \wedge x \notin C \\
&\Leftrightarrow (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin C) \\
&\Leftrightarrow (x \in A \setminus B) \wedge (x \in A \setminus C) \\
&\Leftrightarrow x \in (A \setminus B) \cap (A \setminus C).
\end{aligned}$$

6. Let $f : \mathcal{P}(\mathbb{N}) \to \mathbb{N}$ be the function defined as follows:

$$f(S) = \begin{cases} \min(S) & \text{if } S \text{ has at least 2 elements,} \\ |S| & \text{otherwise.} \end{cases}$$

Here as usual $|S|$ denotes the number of elements of S.

(a) **[1 point]** Is f a surjection? You should justify your answer.

Solution: Yes, f is a surjection because for every $n \in \mathbb{N}$ we have $f(\{n, n+1\}) = n$.

(b) **[3 points]** Compute $\bigcap_{S \in \mathcal{U}} f(\{S, \varnothing\})$ where $\mathcal{U} := \{A \in \mathcal{P}(X) : 0 \notin A \text{ and } 1 \in A\}$. You should justify your answer.

Solution: We have $f(\varnothing) = 0$; further $\varnothing \notin \mathcal{U}$. Now take any $S \in \mathcal{U}$. There are two cases:

- if $|S| = 1$ then we have $f(S) = 1$, hence $f(\{S, \varnothing\}) = \{0, 1\}$.
- if S has at least two elements, then since $0 \notin S$ and $1 \in S$ we must have $f(S) = \min S = 1$. It follows that $f(\{S, \varnothing\}) = \{0, 1\}$.

Thus $\bigcap_{S \in \mathcal{U}} f(\{S, \varnothing\}) = \{0, 1\}$.

7. Recall that $[0,1) = \{t \in \mathbb{R} : 0 \leq t < 1\}$. Set $X := [0,1)$ and consider the partial order $\preceq$ on $X \times X$ defined as follows:

$$(x,y) \preceq (x',y') \Leftrightarrow \big((x \leq x') \wedge (y = y')\big).$$

You do not need to verify that $\preceq$ is a partial order.

(a) [**2 points**] Does $(X \times X, \preceq)$ have any minimum elements? If minimum elements exist, you should determine all of them. You should justify your answer.

Solution:
There are no minimum elements: if $(a,b) \in X \times X$ is a minimum element, then we have $b < 1$, hence we can consider a sufficiently small $\epsilon > 0$ such that $b + \epsilon < 1$. Now $(a, b+\epsilon) \in X \times X$ and we have $(a,b) \not\preceq (a, b+\epsilon)$, which contradicts the assumption that $(a,b) = \min X \times X$.

(b) [**2 points**] Does $(X \times X, \preceq)$ have any minimal elements? If minimal elements exist, you should determine all of them. You should justify your answer.

Solution: Yes. Indeed we claim that the minimal elements of $X \times X$ are precisely the pairs $(0,b)$ for $b \in [0,1)$. Indeed if $(a',b') \preceq (0,b)$ then we must have $0 \leq a' \leq 0$ and $b' = b$, hence $(a',b') = (0,b)$. This proves that $(0,b)$ is minimal.

Also, if $(a,b) \in X \times X$ is a minimal element, then we must have $a = 0$; otherwise $a > 0$ and thus for some suffciently small $\epsilon > 0$ we have $a - \epsilon > 0$, hence $(a-\epsilon, b) \in X \times X$ and clearly we have

$$(a-\epsilon, b) \preceq (a,b) \text{ and } (a-\epsilon, b) \neq (a,b).$$

This contradicts minimality of (a,b).

8. (a) **[1 point]** Write down the definition of a countable set.

Solution: We say A is countable if there exists an injection $f : A \to \mathbb{N}$.

(b) **[3 points]** Let X denote the set of real numbers of the form $\dfrac{\sin(t)}{1+s^2t^2}$ where $s, t \in \mathbb{Q}$. Prove that X is countable.

Solution: It suffices to prove that there exists a surjection $\mathbb{N} \to X$. Since we have $\mathbb{Q} \sim \mathbb{N}$, we also have $\mathbb{Q}\times\mathbb{Q} \sim \mathbb{N}\times\mathbb{N} \sim \mathbb{N}$. Thus, it suffices to give a surjection $\mathbb{Q}\times\mathbb{Q} \to X$. Thus can be done as follows: we define the map

$$f : \mathbb{Q} \times \mathbb{Q} \to X \ , \ f(s,t) = \frac{\sin(t)}{1+s^2t^2}.$$

Clearly f is a surjection.

(c) **[3 points]** Prove that for every set X there does not exist a surjection

$$\phi : X \to \mathrm{Func}(X, \mathbb{N}).$$

Solution: If such a surjection exists then it follows that $\mathrm{Func}(X,\mathbb{N}) \leq X$. However, we prove that $X < \mathrm{Func}(X,\mathbb{N})$. Indeed from $\{0,1\} \leq \mathbb{N}$ it follows that

$$\mathrm{Func}(X,\{0,1\}) \leq \mathrm{Func}(X,\mathbb{N}).$$

Furthermore, in class we proved that $\mathrm{Func}(X,\{0,1\}) \sim \mathcal{P}(X)$. Thus by Cantor's theorem we have

$$X < \mathcal{P}(X) \sim \mathrm{Func}(X,\{0,1\}) \leq \mathrm{Func}(X,\mathbb{N}).$$

9. (a) **[2 points]** Write down the definition of a well-ordered set.

Solution: A totally ordered set X is called well-ordered if for every nonempty $S \subseteq X$, the minimum of S exists and belongs to S.

(b) **[2 points]** Let $X = \left\{\frac{n}{n+1} : n \in \mathbb{N}\right\} \cup \{1\}$, equipped with the total order $\leq$ that is naturally induced from $\mathbb{R}$. Is $(X, \leq)$ well-ordered? You should justify your answer.

Solution: Yes, X is well-ordered. Let $S \subseteq X$ be a nonempty subset of X. If $S = \{1\}$ then clearly $\min S = 1 \in S$. Otherwise, S must contains fractions of the form $\frac{n}{n+1}$. Set $n_\circ := \min\{n : \frac{n}{n+1} \in S\}$. This minimum exists because $(\mathbb{N}, \leq)$ is well-ordered. We prove that $\min S = \frac{n_\circ}{n_\circ+1}$. Indeed for every $t \in S$, if $t = 1$ then we have

$$\frac{n_\circ}{n_\circ + 1} < 1 = t,$$

and if $t = \frac{n}{n+1}$ for $n \geq n_\circ$ then we have

$$\frac{n_\circ}{n_\circ + 1} \leq \frac{n}{n+1} \Leftrightarrow n_\circ(n+1) \leq (n_\circ + 1)n \Leftrightarrow n_\circ \leq n.$$

Since the right hand side inequality is true, the left hand side inequality is also true. This implies that $\min S = \frac{n_\circ}{n_\circ+1}$.

10. Consider the universe with 4 distinct sets a, b, c, d with the following inclusion relations:

$$a \in b \quad , \quad b \in c \quad , \quad d \in b \quad , \quad c \in c$$
$$a \in d \quad , \quad b \in d \quad , \quad c \in d \quad , \quad d \in d.$$

(a) **[1 point]** Does Axiom 4 of ZFC hold in this universe? You should justify your answer.

Axiom 4 (Pair): $\forall_x \forall_y \exists_z \forall_t (t \in z \Leftrightarrow t = x \vee t = y)$

Solution: This axiom does not hold. Axiom 4 is the parity axiom, stating that for any two sets x, y, the (informally written) $\{x, y\}$ is also a set. However, this fails for $x = a$ and $y = b$.

(b) **[1 point]** Does Axiom 5 of ZFC hold in this universe? You should justify your answer.

Axiom 5 (Union): $\forall_x \exists_y \forall_z [\exists_t (z \in t \wedge t \in x) \Leftrightarrow (z \in y)]$

Solution: Axiom 5 holds. Indeed informally, we have

$$a = \{\} \ , \ b = \{a, d\} \ , \ c = \{b, c\} \ , \ d = \{a, b, c, d\}$$

Thus in Axiom 5, for $x = a$ we can set $y = a$, for $x = b$ we can set $y = d$, for $x = c$ we can set $y = d$, and for $x = d$ we can set $y = d$.